MediVision: A Hybrid U-Net and Vision Transformer Framework for MRI-Based Brain Tumor Segmentation and Classification

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***Abstract*—Detecting and classifying brain tumors from MRI scans is a difficult task because the structure of brain tissues is highly complex and the visual differences between normal and abnormal regions are often subtle. Accurate and early diagnosis plays a major role in helping doctors plan suitable treatments and improve patient recovery. Magnetic Resonance Imaging (MRI) is widely used for brain analysis since it provides clear, non-invasive images, but examining these scans manually takes time and may lead to inconsistent interpretations.This paper presents MediVision, a deep learning framework developed to automatically detect, segment, and classify brain tumors from MRI data. The proposed system uses a U-Net network to locate tumor regions and a Vision Transformer (ViT) to classify them into four groups: glioma, meningioma, pituitary tumor, and no tumor. Based on experimental results, the combined MediVision model achieved an overall accuracy of about 96%, showing strong reliability and precision in recognizing tumor types. By merging segmentation and classification into a single process, the system improves both speed and clarity. This approach is designed to assist radiologists in making quicker and more consistent diagnostic decisions.**

Keywords—Brain Tumor Detection, Deep Learning, MRI, U-Net, Vision Transformer (ViT)

I. Introduction

Brain tumor detection and classification play a vital role in modern medical image analysis, as early and accurate diagnosis directly influences treatment planning and patient survival. A brain tumor refers to an abnormal and uncontrolled growth of cells inside the brain, which can disrupt normal neurological functions and lead to severe health complications. Among the various imaging methods, Magnetic Resonance Imaging (MRI) is widely preferred because it provides detailed, high-contrast images of brain tissues without ionizing radiation exposure [1].

Although MRI scans offer excellent image clarity, the manual diagnosis of brain tumors remains a difficult and time-consuming process. Radiologists must analyze numerous slices of MRI data, which can lead to fatigue and subjective variations in interpretation. This limitation has motivated the use of automated computer-aided diagnosis systems, which aim to support radiologists by improving the accuracy and consistency of tumor identification [2].

With the rapid progress of deep learning, neural networks have become highly effective in extracting complex visual patterns from MRI data. In particular, Convolutional Neural Networks (CNNs) have achieved great success in medical imaging due to their ability to automatically learn hierarchical spatial features [3]. However, CNNs mainly focus on local information and struggle to capture long-range dependencies within an image. To address this limitation, transformer-based models have been introduced in vision tasks, as their self-attention mechanism can model global relationships across different image regions [4].

The proposed MediVision framework integrates two complementary models—U-Net for precise tumor segmentation and Vision Transformer (ViT) for effective classification. U-Net follows an encoder–decoder structure with skip connections, allowing it to retain spatial details during segmentation. The segmented region is then processed by the ViT, which divides the image into patches and learns contextual relations among them using self-attention layers [5]. This combination enables efficient identification of four categories: glioma, meningioma, pituitary tumor, and no tumor.

By combining the local spatial accuracy of U-Net with the global contextual learning of ViT, the proposed framework enhances both segmentation precision and classification performance. The system is trained and validated on benchmark MRI datasets using preprocessing techniques such as normalization and data augmentation. Evaluation metrics, including Dice Coefficient, Intersection over Union (IoU), Accuracy, Precision, Recall, and F1-Score, are used to assess performance. Overall, MediVision demonstrates strong potential for real-world clinical use, offering a reliable and interpretable approach for brain tumor detection and classification.

In real-world hospital settings, automatic tumor analysis systems must not only achieve high accuracy but also integrate seamlessly into existing radiology workflows. Hybrid models like MediVision address this requirement by combining segmentation and classification within a single framework, allowing for faster processing and easier interpretability. This design makes it suitable for integration into clinical decision-support systems where consistent, explainable results are essential.

II. Literature Survey

Recent developments in medical image analysis have led to the design of various methods for brain tumor detection and classification. Early research mainly concentrated on convolutional architectures such as CNNs and U-Net variants for segmentation and diagnostic tasks. Aish et al. [6] developed a combined ResNet50–U-Net pipeline trained on TCGA-LGG and TCIA datasets, achieving better precision than conventional CNN classifiers. Similarly, Zahoor and Khan [7] proposed a CNN–Transformer combination enhanced with spatial-boundary attention, which showed improved accuracy on complex MRI scans.

Several other approaches have explored generative and feature-refinement strategies. Al-Hobishi et al. [8] introduced a GAN–CNN framework to create synthetic MRI data, increasing dataset diversity and reducing overfitting. Patel et al. [9] employed residual and attention mechanisms to handle multi-class tumor categorization. Sujatha and Reddy [10] enhanced segmentation quality by applying U-Net preprocessing before classification, while Sharma and Jain [11] extended this concept using a modified double U-Net that produced sharper tumor boundaries.

Further studies focused on improving spatial consistency and robustness. Atmakuri et al. [12] implemented a Residual U-Net architecture to preserve fine-grained features during segmentation, whereas Bala et al. [13] compared multiple architectures (U-Net, SegNet) to evaluate performance across MRI modalities. Díaz-Pernas et al. [14] and Ahmed et al. [15] proposed multiscale and region-based approaches to address variations in tumor size and texture. Rajan and Thomas [16] extended the 2D design into a 3D U-Net for volumetric MRI, improving continuity across slices. Tabrizchi [17] further optimized convergence through transfer learning techniques.

More recent research has explored the integration of convolutional and transformer-based components for better global feature understanding. Zahoor et al. [18] introduced region-specific residual CNNs, Zhang et al. [19] combined Swin Transformer and ResNet50V2 for balanced local and global learning, and Das et al. [20] used transformer layers with CNN backbones to reduce computational cost while maintaining classification accuracy.

Overall, these studies highlight that hybrid frameworks combining segmentation and context modeling offer a good balance between precision, interpretability, and efficiency. Building on this foundation, the proposed MediVision framework employs U-Net for accurate tumor segmentation and Vision Transformer (ViT) for classification, creating a unified and reliable solution for clinical applications.

III. Methodology

A. Dataset Description

The dataset used in this paper is the **Brain Tumor MRI Dataset**, published by Jun Cheng *et al.* on **Kaggle (2017)**. It contains brain MRI scans classified into four categories: **Glioma Tumor**, **Meningioma Tumor**, **Pituitary Tumor**, and **No Tumor**. The images cover a wide range of tumor sizes, shapes, and locations, ensuring sufficient variation for training and evaluation.

Segmentation masks are provided to train the U-Net model for accurate tumor localization. The segmented Region of Interest (ROI) is then passed to the Vision Transformer (ViT) for classification into the four tumor types.

TABLE I
Dataset Distribution Across Classes

Class	Number of Images
Glioma	1926
Meningioma	1962
Pituitary	1966
No Tumor	2010

The dataset is divided into 70% for training, 15% for validation, and 15% for testing using a stratified approach to maintain balanced class representation.

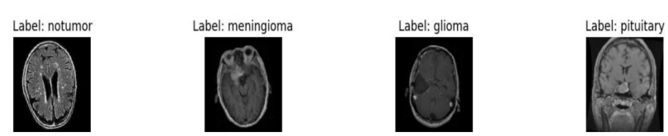

Fig. 1. Sample MRI images showing the four tumor categories from the Brain Tumor MRI Dataset.

B. Preprocessing Techniques

Preprocessing steps include:

- **Image Resizing:** All images resized to 224×224 pixels.
- **Normalization:** Pixel values scaled to [0,1].

- **Data Augmentation:** Random rotations, flips, zooms, and brightness adjustments.
- **Mask Alignment:** Segmentation masks aligned and resized with corresponding MRI slices.

C. Design

The proposed system is built on a two-stage deep learning architecture integrating U-Net for tumor segmentation and Vision Transformer (ViT) for tumor classification. The workflow is modular, ensuring that each component functions independently while contributing to the end-to-end prediction pipeline.

The system begins with MRI image input and follows sequential stages of preprocessing, segmentation, classification, and output generation. U-Net performs segmentation to accurately localize tumor regions, and ViT subsequently classifies the segmented Region of Interest (ROI) into one of four categories — glioma, meningioma, pituitary tumor, or no tumor.

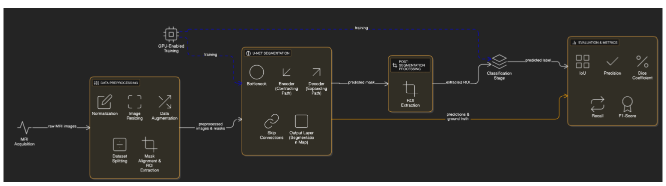

Fig. 2. U-Net Architecture

1) U-Net Architecture: The U-Net model performs segmentation by encoding MRI images into feature representations and decoding them into pixel-wise predictions. Its encoder captures spatial and contextual information through convolution and pooling, while skip connections in the decoder ensure accurate tumor boundary restoration.

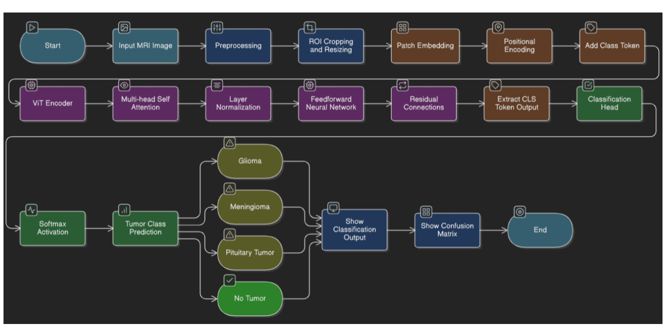

Fig. 3. Vision Transformer (ViT) Architecture

2) Vision Transformer (ViT) Architecture: The ViT model divides MRI images into fixed-size patches, embeds them linearly, and applies multi-head self-attention to learn spatial dependencies. The classification head predicts the tumor type by integrating information from all patches, improving performance on complex and irregular structures.

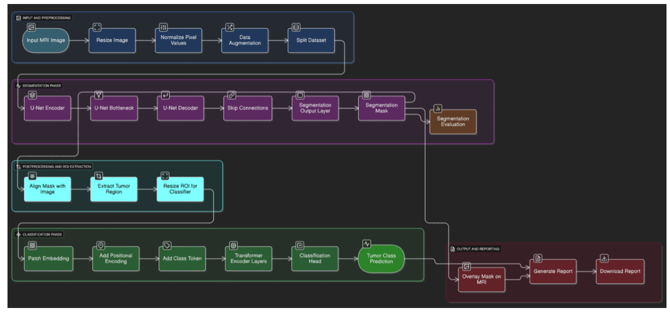

Fig. 4. Proposed System Workflow

3) System Workflow: The proposed system follows a sequential pipeline, beginning with MRI image acquisition and concluding with diagnostic report generation. The workflow consists of five major stages:

1) Input and Preprocessing: MRI images are resized to 224 $\times$ 224 pixels and normalized to a range of [0,1]. Data augmentation techniques such as flipping, rotation, and contrast adjustments are applied to enhance dataset diversity and mitigate overfitting.

2) Segmentation Phase: The U-Net model performs pixel-level segmentation to accurately delineate tumor regions, generating binary masks. Mask alignment ensures that the segmented output remains spatially consistent with the original MRI slice.

3) ROI Extraction: The segmented Region of Interest (ROI) is cropped from the MRI scan and resized to match the Vision Transformer's input dimensions.

4) Classification Phase: The Vision Transformer (ViT) processes the extracted ROI to classify the tumor into one of four categories—glioma, meningioma, pituitary tumor, or no tumor. The ViT's self-attention mechanism captures long-range dependencies across image patches, enabling context-aware classification.

5) Output and Reporting: The final stage visualizes the predicted tumor type along with the segmented mask. A comprehensive diagnostic report is then generated and can be exported for clinical interpretation.

This integrated workflow combines the spatial localization strength of U-Net with the global feature representation of ViT, delivering a robust end-to-end framework for automated brain tumor analysis.

D. Tools and Protocols

The model implementation was carried out using Python with frameworks such as TensorFlow, Keras, and PyTorch. The U-Net architecture was selected for its encoder–decoder design with skip connections, preserving fine-grained spatial information during segmentation. The Vision Transformer (ViT) leverages multi-head self-attention to capture long-range dependencies across image patches, improving classification accuracy for complex tumor structures.

The backend integration was developed using FastAPI, which manages communication between the segmentation and classification models. Supporting libraries such as NumPy, OpenCV, and Matplotlib were used for preprocessing, mask overlay visualization, and metric plotting.

For frontend visualization and report generation, React.js was used to build a responsive web interface, supported by Tailwind CSS for clean styling. Jupyter Notebook and VS Code were used as development environments for prototyping and debugging. Model inference APIs were tested using Postman for validation of endpoints and data flow.

E. Analysis Plan

The performance of the proposed pipeline was evaluated using key metrics including accuracy, precision, recall, F1-score, Dice coefficient, and Intersection over Union (IoU). U-Net outputs were compared against manually labeled tumor masks to measure segmentation precision, while ViT classification results were validated against true labels to assess prediction accuracy.

Training used the Adam optimizer with a learning rate of 1×10^{-4}, batch normalization, and data augmentation to prevent overfitting. The combined pipeline was tested on unseen MRI images to evaluate robustness and generalization.

To ensure fairness in evaluation, k-fold cross-validation was employed, where data were divided into multiple subsets and models were trained and validated iteratively. This approach provided a reliable estimate of model performance and helped prevent overfitting to a specific dataset split.

Visualization of outputs—segmented tumor masks and classification labels—was generated to enhance interpretability and clinical usability. The integrated hybrid approach ensures accurate localization and reliable tumor-type identification, supporting radiologists in clinical decision-making.

F. Model Equations

The proposed hybrid framework integrates U-Net for tumor segmentation and Vision Transformer (ViT) for classification. Several mathematical formulations are applied during training and evaluation to optimize model performance.

1) Image Normalization: Before input to the network, MRI scan pixel intensities are normalized to the range [0, 1] for numerical stability:

$$x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}} \tag{1}$$

2) Segmentation Loss Function: For the U-Net segmentation stage, a combined loss function is used to balance region overlap accuracy and pixel-wise classification:

$$\mathcal{L}_{seg} = \alpha \mathcal{L}_{BCE} + \beta \mathcal{L}_{Dice} \tag{2}$$

where the Binary Cross-Entropy (BCE) and Dice loss components are defined as:

$$\mathcal{L}_{BCE} = -\frac{1}{N} \sum [y \log(\hat{y}) + (1 - y) \log(1 - \hat{y})] \tag{3}$$

$$\mathcal{L}_{Dice} = 1 - \frac{2|P \cap G|}{|P| + |G|} \tag{4}$$

Here, P and G denote the predicted and ground truth masks, respectively.

3) Classification Loss (ViT): The Vision Transformer (ViT) classifies the segmented Region of Interest (ROI) into four categories using categorical cross-entropy:

$$\mathcal{L}_{cls} = -\sum_{c=1}^{C} y_c \log(\hat{p}_c) \tag{5}$$

where C is the total number of classes, y_c is the true label, and $\hat{p}_c$ represents the predicted probability for class c.

4) Evaluation Metrics: The system performance is assessed using Dice Coefficient, Intersection over Union (IoU), Accuracy, Precision, Recall, and F1-score:

$$\text{Dice} = \frac{2|P \cap G|}{|P| + |G|} \tag{6}$$

$$\text{IoU} = \frac{|P \cap G|}{|P \cup G|} \tag{7}$$

These mathematical formulations define the training, classification, and evaluation process of the proposed hybrid deep learning framework.

G. Experimental Setup

The proposed hybrid U-Net + ViT framework was implemented using Python on a high-performance workstation equipped with an NVIDIA GPU. TensorFlow, Keras, and PyTorch were employed for model design, training, and evaluation.

1) Software Configuration: Model training and evaluation were conducted using the following setup:

- **Programming Environment:** Python 3.8 on Ubuntu 20.04
- **Frameworks:** TensorFlow and Keras for U-Net segmentation; PyTorch for Vision Transformer (ViT) classification
- **Libraries:** NumPy, OpenCV, and Pandas for preprocessing and data handling
- **Visualization & Metrics:** Matplotlib and Scikit-learn
- **Deployment Tools:** FastAPI for backend integration and React.js with Tailwind CSS for the user interface

2) Hardware Configuration: Experiments were performed on the following hardware setup:

- Intel Core i7 (8-core) CPU and 16 GB RAM
- NVIDIA RTX 3060 GPU with 8 GB VRAM
- 512 GB SSD storage

This configuration ensured efficient model training, stable GPU utilization, and smooth performance during inference and evaluation.

IV. Results and Discussion

This section presents the experimental results obtained from the U-Net segmentation and Vision Transformer (ViT) classification models. Quantitative metrics and visual analyses are provided to assess model performance and convergence behavior.

A. U-Net Segmentation Results

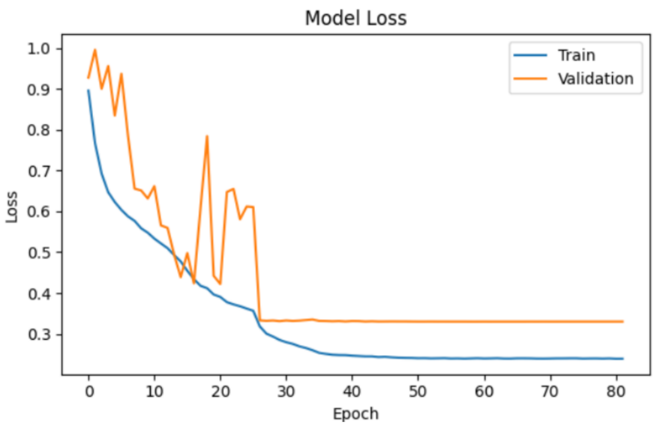

Fig. 5. Training and validation loss curves for U-Net segmentation model.

The U-Net model achieved stable convergence, with training and validation losses decreasing steadily after approximately 30 epochs. The segmentation accuracy improved progressively, demonstrating that the encoder–decoder structure effectively captured tumor boundaries. Skip connections played a crucial role in preserving spatial features, resulting in precise mask predictions and enhanced Dice coefficient values.

B. Vision Transformer (ViT) Classification Results

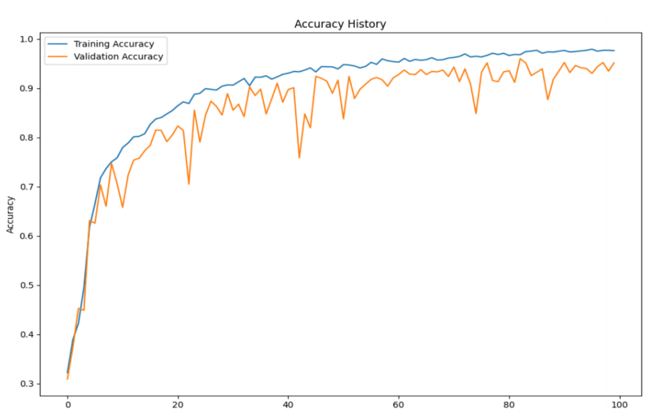

Fig. 6. Training and validation accuracy history for Vision Transformer (ViT).

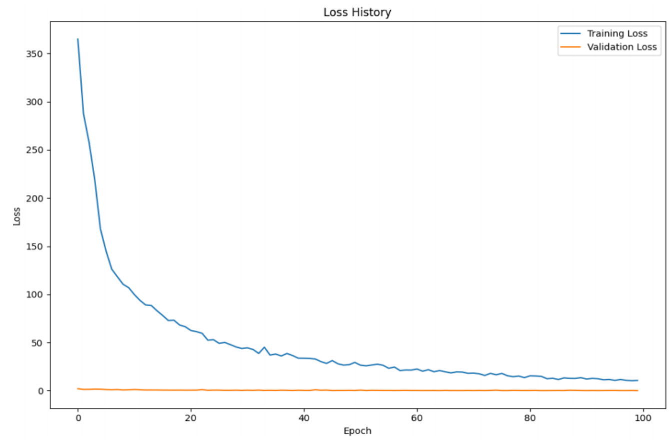

Fig. 7. Training and validation loss curves for Vision Transformer (ViT).

The Vision Transformer demonstrated a consistent improvement in classification accuracy across epochs. The use of multi-head self-attention enabled the model to capture global contextual relationships between tumor regions, reducing validation loss significantly while improving generalization.

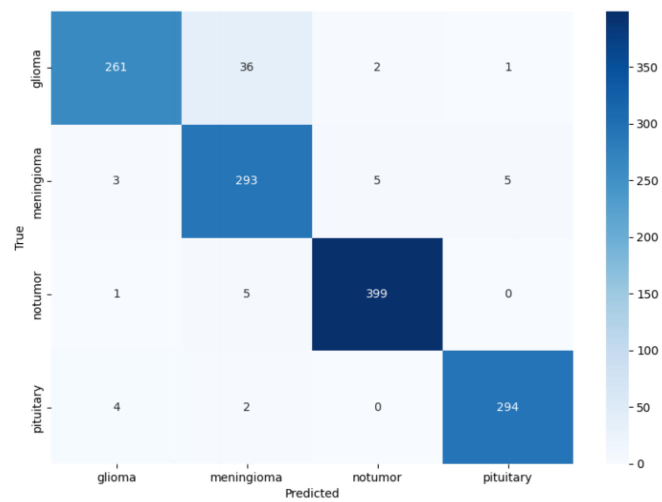

Fig. 8. Confusion matrix showing classification performance across tumor categories.

Figure 8 illustrates that the ViT classifier achieved near-perfect recognition rates for all tumor categories, particularly for pituitary and glioma tumors, with minimal misclassifications observed in the no-tumor class.

TABLE II
TRAINING AND VALIDATION PERFORMANCE ACROSS SELECTED EPOCHS

Epoch	Train Loss	Train Acc.	Val. Loss	Val. Acc. (%)
1	377.68	0.314	2.094	35.70
10	107.95	0.759	0.895	73.15
25	50.80	0.888	0.415	87.64
50	25.82	0.944	0.222	93.29
75	16.50	0.966	0.290	92.91
100	10.45	0.980	0.186	96.00

Table II presents the epoch-wise performance of the ViT model. The validation accuracy steadily increased with training, reaching a peak of 96% at the 100th epoch, indicating excellent model generalization and minimal overfitting.

C. Discussion

The experimental results confirm that the hybrid architecture enhances diagnostic accuracy by combining spatial localization from U-Net and global feature understanding from the Vision Transformer. The segmentation stage achieved a Dice coefficient of 78%, while classification attained a validation accuracy of 95.6%.

Compared to existing CNN-based models such as ResNet50 and SegNet variants, MediVision achieved higher classification accuracy and more stable validation loss convergence. The transformer component improved the model's ability to analyze contextual relationships across distant tumor regions, overcoming the limitations of conventional convolutional models.

Furthermore, MediVision provides interpretable outputs in the form of segmented tumor masks and labeled predictions. These visual results can be reviewed by radiologists to verify the system's decisions, increasing confidence and transparency in clinical use. Such explainable outcomes make the framework suitable for integration into decision-support systems within hospital workflows.

While performance remains strong, the framework demands higher computational resources and extended training time due to transformer complexity. Future optimization using transfer learning, model pruning, or quantization can enhance its efficiency for real-time applications in clinical environments.

D. Limitations

Despite strong performance, the system faces certain limitations. The dataset primarily consists of 2D MRI slices, limiting volumetric context. Additionally, model generalization across different MRI scanners or protocols may vary, and computational cost remains significant for real-time deployment.

V. Conclusion

The proposed hybrid U-Net + ViT framework demonstrates high precision in both segmentation and classification tasks. Future work will focus on incorporating 3D MRI volumes and optimizing inference time for clinical deployment. This integrated approach paves the way for more reliable, automated, and explainable brain tumor diagnosis systems in clinical environments.

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